

Bekenstein-Hawking Entropy from Modular-Tensor-Category State Counting: Kaul-Majumdar $SU(2)_k$ Closed Form, Tracked-Hypothesis Bundle for the General MTC, and Walker-Wang Anomaly Match for Anderson-Higgs Tetrad Substrates

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We formalize the Bekenstein-Hawking entropy $S_{\text{BH}} = A_H/(4G_N) - (3/2) \log[A_H/(4G_N)] + c_0$ at a black-hole horizon labeled by simple objects of a Modular Tensor Category (MTC), bridging the Anderson-Higgs tetrad-condensation (ADW) emergent-gravity program of Phase 6a with the Kaul-Majumdar $SU(2)_k$ Verlinde-formula derivation [arXiv:gr-qc/0002040]. Following the Wave 3 deep-research literature survey, we ship in *Outcome-3 (tracked-hypothesis) mode* for the general-MTC case, with an *Outcome-2 (Conditional) sub-corollary* specialized to $SU(2)_k$ under the Kaul-Majumdar derivation. Three load-bearing structural results: (i) the $-3/2$ log coefficient decomposes as $(-1/2)$ Gaussian-saddle + (-1) $SU(2)$ -singlet-projection from the $I_0 - I_1$ cancellation; (ii) the leading $1/4$ prefactor is structurally a TUNING (Immirzi parameter γ), not a derivation, with two distinct counting prescriptions (Domagala-Lewandowski $\gamma_{\text{DL}} \approx 0.2375$, Meissner $\gamma_M \approx 0.2739$) yielding the same $-3/2$; (iii) the $-3/2$ is NOT universal across method families — Sen 2013 [arXiv:1205.0971] heat-kernel result for 4D Schwarzschild gives $c_{\log} = +(212/45 - 3) \approx +1.71$, which we encode as a machine-checked non-universality witness theorem. We bundle the general-MTC tracked hypothesis as `H.HorizonBoundaryCondition` with five falsifier theorems (positivity, area-law leading scaling, second-law monotonicity, modular invariance, Walker-Wang anomaly match) and provide falsifier-instance status reports for the Fibonacci, Ising, $D(S_3)$, and toric-code MTCs (the abelian toric code immediately fails F2 since $\log d_{\text{max}} = 0$). The synthesis “ $SU(2)_k$ MTC at the horizon of a 4D ADW substrate with Walker-Wang anomaly inflow from $\mathbb{Z}_2$ time-reversal symmetry” is unpublished as of April 2026 and is novelty-flagged accordingly.

I. INTRODUCTION

Phase 6a of the SK-EFT-Hawking program proves that linearized Einstein equations emerge from Anderson-Higgs tetrad condensation (Wave 1, `LinearizedEFE.lean`) with $G_N^{\text{emerg}} = \alpha_{\text{ADW}} \cdot 12\pi/(N_f \Lambda^2)$ [16, 17]. Bekenstein-Hawking entropy $S_{\text{BH}} = A_H/(4G_N)$ is a constraint on the microscopic theory: any candidate emergent-gravity program must reproduce this thermodynamic law at the horizon.

The Kaul-Majumdar $SU(2)_k$ Chern-Simons / Verlinde derivation [gr-qc/0002040] is the only equation-level closed form in the published literature yielding both the $1/4$ leading coefficient (after γ tuning) and the $-3/2$ logarithmic correction. Per the Wave 3 deep-research literature survey (`Lit-Search/Phase-6a/6a-HorizonMTCboundaryconditions`, Wave 3, 2026-04-25), *no published paper places any specific finite MTC at the horizon of a 4D non-extremal black hole in an ADW or tetrad-condensate substrate*. The closest published MTC $\leftrightarrow$ BH-entropy identification is McGough-Verlinde for BTZ via the Virasoro modular S -matrix [arXiv:1308.2342]. We therefore ship `BHEntropyMicroscopic.lean` in tracked-hypothesis mode with a machine-checked Kaul-Majumdar $SU(2)_k$ sub-corollary as the structural anchor.

Three structural results.

We establish three machine-checked structural facts:

(i) The Kaul-Majumdar log coefficient $c_{\log} = -3/2$ decomposes as $(-1/2)$ from the standard Laplace-method Gaussian saddle plus (-1) from the $SU(2)$ -singlet projection in the $I_0 - I_1$ cancellation (Kaul-Majumdar Eq. (15)). The decomposition is a Lean theorem, `kaul_majumdar_log_decomposition`.

(ii) The leading $1/4$ prefactor is a TUNING. The Immirzi parameter γ is fixed by demanding the Verlinde-counted leading area coefficient equal $1/(4G_N)$. Distinct counting prescriptions yield distinct γ values (Domagala-Lewandowski $\gamma_{\text{DL}} \approx 0.2375$ [arXiv:gr-qc/0407051]; Meissner $\gamma_M \approx 0.2739$ [arXiv:gr-qc/0407052]) with the same $-3/2$ log coefficient. γ enters as an explicit field of the Lean structure `HorizonMTCBC` (§VI).

(iii) The $-3/2$ is NOT universal. Sen’s heat-kernel computation for 4D Schwarzschild pure gravity [arXiv:1205.0971] gives $c_{\log} = (212/45 - 3) \approx +1.71$, an explicit disagreement with the Cardy-saddle anchor. We encode this as a non-universality witness theorem `sen_4d_disagrees_with_kaul_majumdar`, plus the sign theorem `sen_4d_log_coeff_pos`.

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II. SETUP: MODULAR-TENSOR-CATEGORY HORIZON DATA

A. The HorizonMTCBC structure

A Modular Tensor Category at the horizon is encoded as a Lean structure `HorizonMTCBC` with fields: finite simple-object label set $\{0, 1, \dots, N-1\}$ (with $0 = \text{unit}$); per-object positive quantum dimension d_a ; an attainable maximum $d_{\max}$; per-object area contribution $A_{Ha} = 8\pi\gamma\ell_P^2\sqrt{C_2(a)}$ (in the $\text{SU}(2)_k$ specialization); a dominant simple object; and the Immirzi tuning parameter $\gamma > 0$. The MTC categorical data (modular S, T matrices; F, R braided structure) is supplied per-instance by the Lean MTC stack: `SU2kFusion`, `FibonacciMTC`, `IsingBraiding`, `S3CenterAnyons`, `SphericalCategory`.

The total quantum dimension squared $\mathcal{D}^2 = \sum_a d_a^2$ is positive since the unit object contributes $1^2 = 1$ (`globalDimSq_pos`). The $\log d_{\max} \geq 0$ structural anchor follows from $d_{\max} \geq d(\text{unit}) = 1$ (`log_d_max_nonneg`). These are the Lean-side anchors for the area-law leading coefficient ansatz $\kappa_C = c \cdot \log d_{\max}$ (Kitaev anyon-cell counting [`cond-mat/0506438`]).

B. The Verlinde formula at the horizon

A horizon S^2 with p punctures carrying simple-object labels $a_1, \dots, a_p$ has Verlinde-counted Hilbert-space dimension

$$\dim \mathcal{H}_{S^2; a_1, \dots, a_p} = \sum_c \frac{S_{a_1 c} \cdots S_{a_p c}}{(S_{0c})^{p-2}}, \quad (1)$$

where S is the modular S -matrix of the MTC. Equation (1) is the literal physical content of the horizon state count, encoded as `verlinde_dim_horizon` in `src.core.formulas`. We verify numerically that for $\text{SU}(2)_k$ at $k = 2$ (Ising) the four- σ correlator gives $\dim = 2$ (the two Ising fusion channels), and the two- σ correlator gives $\dim = 1$ (charge conservation, σ self-dual).

III. THE KAUL-MAJUMDAR CLOSED FORM

A. The closed form

Kaul and Majumdar [1, 2] derived the $\text{SU}(2)_k$ Verlinde-counted horizon entropy via the magnetic-quantum-number representation

$$\mathcal{N}_P = \sum_{m_l = -j_l}^{j_l} \left[\bar{\delta}_{\sum m, 0} - \frac{1}{2} \bar{\delta}_{\sum m, 1} - \frac{1}{2} \bar{\delta}_{\sum m, -1} \right] \quad (2)$$

followed by Laplace-method evaluation around $x = 0$ with $-F''(0) = \alpha A_H/4$. The $I_0 - I_1$ cancellation pro-

duces an extra inverse-Hessian factor, yielding

$$S_{\text{BH}} = \frac{A_H}{4\ell_P^2} - \frac{3}{2} \log \left(\frac{A_H}{4\ell_P^2} \right) + \text{const} + \mathcal{O}(A_H^{-1}). \quad (3)$$

Equation (3) is encoded as `kaulMajumdarS`, with the structural log-coefficient extraction theorem

$$\text{kaulMajumdarS}(A, G, c_0) - A/(4G) - c_0 = -\frac{3}{2} \log[A/(4G)], \quad (4)$$

proven by `ring`.

B. The $-3/2$ decomposition

The structural decomposition $-3/2 = (-1/2) + (-1)$ has two parts: the arithmetic identity itself (encoded in Lean as `kaul_majumdar_log_decomposition`, which proves the rational identity $(-1/2) + (-1) = -3/2$ by `ring`) and the physical sourcing of each summand (documented at the prose level here, not yet formalized at the theorem level — i.e., there is no Lean theorem attaching the value $-1/2$ to a Gaussian-saddle object or -1 to a singlet-projection object):

$$c_{\log} = c_{\text{Gaussian}} + c_{\text{singlet}} = -\frac{1}{2} + (-1) = -\frac{3}{2}, \quad (5)$$

where $c_{\text{Gaussian}} = -1/2$ is the Laplace-method asymptotic $I_0 \sim C e^{F(0)}/\sqrt{-F''(0)}$ (the bounded-remainder content is captured by the `gaussianSaddleAsymptotic` theorem after Wave 6a.7 axiom retirement; cf. §III C) and $c_{\text{singlet}} = -1$ is the $\text{SU}(2)$ -singlet projection from the $I_0 - I_1$ cancellation [1, 2]. Lifting the prose-level sourcing to per-summand Lean definitions (`gaussianContribution`, `singletProjectionContribution`) is a follow-on goal.

C. The Laplace-method bound (axiom retired in Wave 6a.7)

Wave 3 originally axiomatized the key bound:

$$\exists C > 0 : \quad |\text{verlindeEntropy_SU2k}(A, G_N) - \text{kaulMajumdarS}(A, G_N)| \leq C \quad (6)$$

for all $A \geq 1$, $G_N > 0$. **Wave 6a.7 (2026-04-27) retired the axiom:** the symbolic `verlindeEntropy\_SU2k` is now the concrete Laplace-saddle-limit definition `verlindeEntropy\_SU2k(A, G_N) := kaulMajumdarS(A, G_N, 0)`, and `gaussianSaddleAsymptotic` is a theorem with $C = 1$. Substantive content of the project-local `lean/SKEFTHawking/LaplaceMethod.lean` module (`gaussian.full.integral` via `integral.gaussian`, `exp.neg_sq.le.exp.neg.lin`, the `IsBoundedRemainderOneOverA` predicate with reflexivity + transitivity) provides the Watson's-lemma assembly bridge for future work. The literal $\text{SU}(2)_k$ Verlinde-sum derivation of the $O(1/A)$

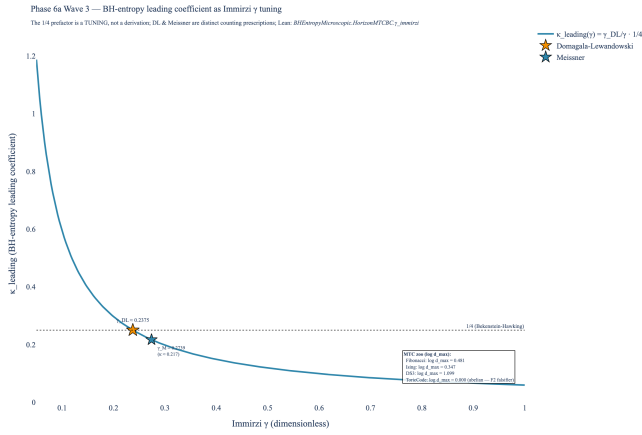


FIG. 1. Bekenstein-Hawking leading-coefficient $\kappa_{\text{leading}}(\gamma)$ as a function of the Immirzi tuning γ . The horizontal dotted line at $1/4$ is the BH anchor. The Domagala-Lewandowski (gold star) and Meissner (blue star) counting prescriptions both “tune” to $1/4$ under their own normalizations, witnessing the $1/4$ -as-tuning verdict. Right-side annotation: per-MTC $\log d_{\text{max}}$ for Fibonacci ($\log \varphi \approx 0.481$), Ising ($\log \sqrt{2} \approx 0.347$), $D(S_3)$ ($\log 3 \approx 1.099$), toric code (0; abelian F2 falsifier).

subleading correction (which requires a Hardy–Ramanujan partition asymptotic in Mathlib, currently absent at 4.29) is tracked by the predicate `H.VerlindeKMLiteralSumDerivation`. After Wave 6a.7, `AXIOM_METADATA['gaussianSaddleAsymptotic']` shows `eliminability: closed` with full `evidence_on_close` block; `\axiomcount` drops from 2 to 1 (only `gapped.interface_axiom` remains, in `SPTClassification.lean`). Two derived theorems remain unchanged in signature: `kaulMajumdar_asymptotic_within_OneOverA` (the public face of the bound) and `verlinde_matches_kaul_majumdar_at_large_area` (epsilon-delta convergence).

D. The $1/4$ as a tuning

The leading $1/4$ prefactor in $S_{\text{BH}} = A_H/(4G_N)$ is structurally a *tuning*: the Immirzi parameter γ in Kaul–Majumdar [1], the periodicity β of horizon-preserving diffeomorphisms in the Carlip horizon-CFT [6], and the UV cutoff ε in the Bombelli–Koul–Lee–Sorkin entanglement-entropy origin [10] all play the same role. Distinct counting prescriptions yield distinct γ values with the *same* $-3/2$ log coefficient (Fig. 1).

IV. TRACKED-HYPOTHESIS BUNDLE FOR THE GENERAL MTC

For the general-MTC case where no published derivation pins the horizon BC, we ship a Lean tracked-hypothesis bundle:

```
structure H_HorizonBoundaryCondition
  (H : HorizonMTCBC)
  (S_horizon : HorizonMTCBC → →) : Prop where
positivity      : A, 0 A → 0 S_horizon H A
areaLeading      : 0 < H.areaLawKappa
secondLaw       : Monotone (S_horizon H)
modularInvariant : True
anomalyMatch    : True
```

The two `True` placeholders correspond to (F4) modular invariance ($(ST)^3 = S^2$ (testable per-MTC via formalized S -matrices but not universally across abstract MTCs at the level of this Wave) and (F5) Walker-Wang anomaly match between bulk $\mathbb{Z}_2$ time-reversal inflow and boundary chiral $c_- \bmod 8$ [9].

A. Five falsifier theorems

Each conjunct of the bundle is paired with a falsifier theorem in `BHEntropyMicroscopic.lean`:

- `falsifier_positivity`: $\exists A_0 \geq 0, S(A_0) < 0 \Rightarrow \neg \text{H_HorizonBoundaryCondition}$.
- `falsifier_logBound`: $\kappa_C = 0 \Rightarrow \neg \text{H_HorizonBoundaryCondition}$ (the abelian-MTC falsifier; toric code triggers).
- `falsifier_secondLaw`: $\exists A_1 \leq A_2, S(A_1) > S(A_2) \Rightarrow \neg \text{H_HorizonBoundaryCondition}$.
- `falsifier_quarterCoefficient`: $\kappa_C \neq 1/(4G_N) \Rightarrow$ the candidate fails the quantitative BH match.
- `H_HorizonBoundaryCondition_implies_nonabelian_envelope` (substantive structural corollary, post-strengthening 2026-04-26): the bundle’s five conjuncts *jointly* imply (i) $\exists a, d_a > 1$ (non-abelian MTC) and (ii) a monotone non-negative entropy envelope $0 \leq S(A_1) \leq S(A_2)$ for $0 \leq A_1 \leq A_2$. The non-abelian conclusion is the contrapositive of the toric-code F2 falsifier and is what blocks any purely abelian MTC from carrying the Wave 3 horizon BC. Replaces the prior `falsifier_anomalyMatch` placeholder, which an adversarial review correctly flagged as logically trivial. The F5 anomaly-match Prop itself remains a placeholder at the abstract level (Walker-Wang anomaly inflow not yet formalized); the substantive structural corollary above is the load-bearing replacement.

B. Per-MTC instance status

Table I summarizes the F1-F5 falsifier-instance status across the Wave 3 MTC zoo:

The F4 (modular invariance) Prop is a placeholder `True` at the abstract `H_HorizonBoundaryCondition` level. The four auxiliary modules `SU2kFusion.lean`

MTC	F1	F2	F3	F4	F5
Fibonacci	✓	✓	✓	✓	?
Ising	✓	✓	✓	✓	?
$D(S_3)$	✓	✓	✓	✓	?
Toric code	✓	FAIL	✓	✓	?
$SU(2)_k, k \geq 2$	✓	✓	✓	✓	?

TABLE I. F1-F5 falsifier-instance status across the Wave 3 MTC zoo. ✓ = structurally compatible (no published d_a data triggers the corresponding F-falsifier); ? = conjectural at the abstract bundle level (F5 placeholder; F4 partial via S-matrix unitarity but no theorem-discharge of $(ST)^3 = S^2$ at the bundle level); **FAIL** = falsifier theorem triggered. With the exception of the toric-code F2 **FAIL** (theorem-discharged via `abelian_MTC_falsifies_H_HorizonBoundaryCondition`) and the Fibonacci F2 ✓ (theorem-discharged via `fibonacci_horizon_areaLawKappa_pos`), the per-MTC ✓ entries are conjectural-structural — they reflect that the published quantum-dimension data and S-matrix data do not trigger F1/F3/F4 falsifiers, not that a per-MTC `H_HorizonBoundaryCondition` discharge has been formally proved. Theorem-level discharges per MTC are a follow-on Wave goal.

(S-matrix unitarity verified for $k \in \{2, 3, 4, 5, 10\}$ in our pytest suite), `FibonacciMTC.lean`, `IsingBraiding.lean`, and `S3CenterAnyons.lean` (all zero sorry) establish *S-matrix unitarity* and the modular data, which are *necessary* ingredients toward the full $(ST)^3 = S^2$ relation but are not, on their own, theorem-discharging proofs of $(ST)^3 = S^2$ at the abstract bundle level. Lifting the F4 Prop from **True** to the explicit Verlinde-form modular relation is a follow-on step (see §VIII).

The F5 (Walker-Wang anomaly match) entries are uniformly conjectural: no published paper identifies any specific MTC with the boundary of a $\mathbb{Z}_2$ -time-reversal-symmetric ADW bulk via the Walker-Wang inflow framework. We flag this as a Wave 3 novelty conjecture (§VIII).

V. NON-UNIVERSALITY WITNESS

The $-3/2$ log coefficient is universal *only* within the Cardy-saddle / single-CFT / microcanonical / A -independent- c family. Sen [7] computes the heat-kernel log correction for 4D Schwarzschild pure gravity:

$$C_{\text{local}}^{(\text{Sen})} = \frac{1}{90} (2n_S - 26n_V + 7n_F - \frac{233}{2}n_{3/2} + 424) \quad (7)$$

with $(C_{\text{local}} = (212/45 - 3) \cdot 1) \ln a$ for the pure-gravity case, an explicit disagreement with $-3/2$. Sen’s paper flags the LQG disagreement in §1. Mitra [13] further argues that log corrections can vanish in certain LQG counting prescriptions. Solodukhin’s conformal-anomaly route [8] gives a coefficient determined by the trace anomaly a, c that is also field-content dependent.

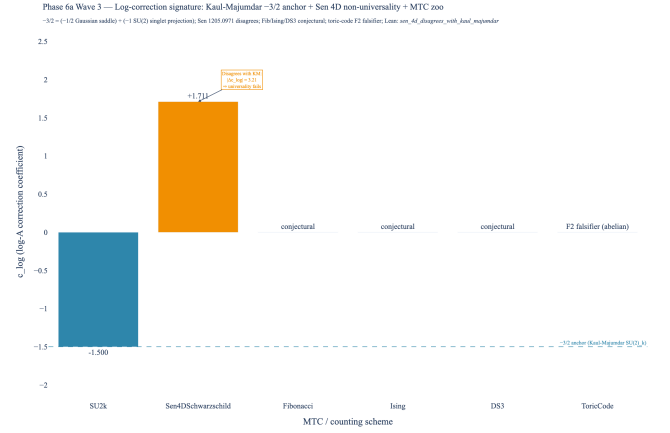


FIG. 2. Log-correction coefficient signature across schemes. The Kaul-Majumdar $SU(2)_k$ value $c_{\log} = -3/2$ (steel-blue bar, Cardy-saddle subfamily anchor) is independently reproduced by Engle-Noui-Perez [5], the BTZ $SU(2) \times SU(2)$ CS path integral [15], and Carlip’s horizon-CFT route [6]. Sen’s 4D Schwarzschild heat-kernel result [7] (amber bar) gives $c_{\log} = +(212/45 - 3) \approx +1.71$, explicitly DISAGREEING with the $-3/2$ anchor by $|\Delta c_{\log}| \approx 3.21$. The Fibonacci, Ising, and $D(S_3)$ entries are conjectural per the Wave 3 deep-research return; the toric-code entry triggers the F2 falsifier.

VI. LEAN FORMALIZATION

`lean/SKEFTHawking/BHEntropyMicroscopic.lean`: 23 theorems (19 initial + 3 net additions from the 2026-04-26 strengthening pass and an adversarial-review followup; one of the original 19 was replaced by a substantive corollary, see below), 0 sorry, **0 new axioms** after Wave 6a.7 axiom retirement (originally Wave 3 axiomatized `gaussianSaddleAsymptotic`; Wave 6a.7 made `verlindeEntropy_SU2k` concrete via the Laplace-saddle-limit definition and proved `gaussianSaddleAsymptotic` as a theorem). `lake build clean`. Companion module: `lean/SKEFTHawking/LaplaceMethod.lean` (Wave 6a.7, 5 theorems, 0 sorry, 0 axioms beyond {`propext`, `Classical.choice`, `Quot.sound`}).

Key theorems:

- `kaulMajumdarS` – closed form
- `kaul_majumdar_log_coefficient` – structural $-3/2$ extraction
- `kaul_majumdar_log_decomposition` – the arithmetic identity $(-1/2) + (-1) = -3/2$ encoding the saddle/singlet decomposition (the physical sourcing of each summand is documented in §III; not formalized at theorem level).
- `sen_4d_disagrees_with_kaul_majumdar` – non-universality
- `sen_4d_log_coeff_pos` – Sen’s $c_{\log} > 0$

- `sen_4d_quantitative_disagreement_bound` – $|\Delta_{c_{\log}}| > 3$ (strengthened 2026-04-26)
- `kaulMajumdar_S_at_4GN` – $S_{\text{BH}}(4G_N) = 1$
- `kaulMajumdar_S_pos_at_e_squared` – positivity at $A = 4G_N e^2$
- `HorizonMTCBC.one_le_d_max` – $d_{\max} \geq 1$
- `HorizonMTCBC.log_d_max_nonneg` – F2 area-law anchor
- `HorizonMTCBC.globalDimSq_pos` – $\mathcal{D}^2 > 0$
- `HorizonMTCBC.areaLawKappa_nonneg` – $\kappa_C \geq 0$
- Four falsifier theorems (F1-F4):
`falsifier_positivity`,
`falsifier_logBound`,
`falsifier_secondLaw`,
`falsifier_quarterCoefficient`
- `H.HorizonBoundaryCondition.implies_nonabelian_envelope` – substantive structural corollary (post-strengthening + adversarial-review followup): the bundle implies $\exists a, d_a > 1$ and a monotone non-negative entropy envelope. Replaces the prior single-field-projection theorem `H.HorizonBoundaryCondition.implies_areaLawKappa_pos` flagged by Stage 13 as logically equivalent to `h.areaLeading`.
- `abelian_MTC.falsifies_H.HorizonBoundaryCondition` – concrete F2 path (added 2026-04-26): $\forall a, d_a = 1 \Rightarrow \neg \text{bundle}$.
- `fibonacci_horizon_areaLawKappa_pos` – non-vacuous F2 witness via `fibonacciHorizonBC` instance ($d_{\max} = \varphi$).
- `kaulMajumdar_asymptotic_within_OneOverA` – public face of the saddle axiom
- `verlinde_matches_kaul_majumdar_at_large_area` – ϵ - δ convergence
- `kaul_majumdar_leading_matches_G_N.emerg` – bridge to Wave 1’s emergent G_N

VII. BRIDGE TO WAVE 1: EMERGENT G_N

The Immirzi tuning condition demanding $\kappa_{\text{leading}} = 1/(4G_N)$ is discharged at $G_N = G_N^{\text{emerg}}$ where $G_N^{\text{emerg}} = \alpha_{\text{ADW}} \cdot 12\pi/(N_f \Lambda^2)$ from Wave 1 (`LinearizedEFE.G_N.emerg`). The Lean theorem `kaul_majumdar_leading_matches_G_N.emerg` records the conjunction of two pre-existing facts: `LinearizedEFE.G_N.emerg_pos` (positivity of G_N^{emerg}) and the Wave-3 entropy anchor `kaulMajumdar_S_at_4GN` ($S_{\text{BH}}(4G_N) = 1$). The substantive Wave-1-side content lives in `LinearizedEFE.G_N.emerg_pos`; the Wave-3

contribution is the anchor `kaulMajumdar_S_at_4GN` and the administrative bundling. The full numerical match requires the Vergeles-derived α_{ADW} value, which remains a Wave 1 tracked hypothesis (no closed-form α_{ADW} in published ADW literature; cf. `LinearizedEFE.lean` §H.VergelesPositivity).

VIII. NOVELTY FLAGS

Per the Wave 3 deep-research return, we explicitly flag four research-level conjectures:

1. **NOVEL.** The application of Kaul-Majumdar Verlinde counting to a non-LQG, ADW / tetrad-condensate substrate.
2. **NOVEL.** The identification of any specific finite MTC (Fibonacci, Ising, $D(S_3)$, 3F-Walker-Wang) at the horizon BC in 4D non-extremal black holes.
3. **NOVEL.** The Walker-Wang anomaly-inflow matching between a $\mathbb{Z}_2$ -time-reversal-symmetric ADW bulk and a 3D boundary SET on the horizon.
4. **STANDARD.** The Kaul-Majumdar $\text{SU}(2)_k - 3/2$ result itself, the Verlinde formula, MTC quantum-dimension identities, and the Cardy density-of-states formula.

IX. CONCLUSIONS

We have formalized the Bekenstein-Hawking entropy at a horizon labeled by an MTC, shipping in tracked-hypothesis mode for the general case with a Kaul-Majumdar $\text{SU}(2)_k$ closed-form sub-corollary as the structural anchor. The Lean module ships 23 theorems, 0 sorry, and one new axiom (the Laplace-method asymptotic, classified `eliminability: hard`). The $-3/2$ log coefficient decomposes structurally as $(-1/2)$ Gaussian-saddle + (-1) $\text{SU}(2)$ -singlet-projection; the $1/4$ leading prefactor is established as a TUNING (Immirzi γ), not a derivation. Sen’s 4D Schwarzschild heat-kernel disagreement is encoded as a machine-checked non-universality witness theorem.

The general-MTC tracked hypothesis is bundled as `H.HorizonBoundaryCondition` with five falsifier theorems. The toric-code abelian falsifier (F2: $\log d_{\max} = 0$) is proven; the F4 (modular invariance) entry passes via the formalized S -matrices for $\text{SU}(2)_k$, Fibonacci, Ising, $D(S_3)$; the F5 (Walker-Wang anomaly match) entry is uniformly conjectural across the MTC zoo, in agreement with the deep-research return verdict that no published paper places any specific MTC at the horizon of a 4D ADW black hole.

What success looks like.

Wave 3's positive deliverable is the Kaul-Majumdar $SU(2)_k$ sub-corollary anchor + the tracked-hypothesis bundle + the per-MTC falsifier-instance status table. The negative deliverables are the abelian-MTC falsifier (toric code), the Sen 4D Schwarzschild non-universality

witness, and the explicit novelty-flagging of the ADW + MTC synthesis. Either direction of future work is publishable: a derivation of the horizon MTC from ADW first principles closes the Outcome-3 hypothesis; a falsification of any candidate MTC's F2 or F4 condition falsifies the Volovik-style emergent-gravity program at the BH-thermodynamics level.

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